

Liam Hastings

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Software Engineer — Actively Seeking Full-Time Opportunities

EDUCATION

Carleton University

September 2022 - May 2026

B.S. in Computer Science, AI and ML stream

- **Relevant Courses:** Data Structures & Algorithms, Systems Programming, Machine Learning, Database Management Systems, Java Object-Oriented Programming, Software Engineering, Web Applications, Software Quality Assurance
- **Awards:** Dean's List, Faculty Scholarship, William and Ann Merrill Bursary

TECHNICAL SKILLS

Languages: Java, Python (NumPy, Pandas, Matplotlib), JavaScript, C, C++, SQL, HTML/CSS

Tools and Frameworks: Node.js, React.js, PyTorch, Git, Cucumber, Cypress, VS Code, Linux, IntelliJ

PROJECTS

Restock – Offline Kitchen Inventory iOS App

- Built an offline **React Native + Expo** iOS app (TypeScript, Zustand, **SQLite**; no backend, account, or sync) that keeps kitchen inventory current as a by-product of deciding what to cook: tapping “have it” or “need it” on a recipe’s ingredients updates inventory and an aisle-grouped grocery list; used daily on my iPhone
- Kept all domain logic free of React and native imports so a **106-assertion rules suite** compiles with tsc and runs under plain Node, with no test framework and no mocks
- Built **two-tier recipe import**: injected JavaScript reads schema.org Recipe JSON-LD over the **WebView bridge**, falling back to the **Claude API** only when a page has none; both tiers fail closed so a bad page never half-saves a recipe
- Wrote an **ingredient-line parser** that turns “2 (14-ounce) cans diced tomatoes” into a canonical item plus free-text quantity, handling fractions, ranges, and size adjectives so the catalog avoids near-duplicates

AmIChalked – Crowdsourced Parking Alert App

- Built a full-stack, cross-platform app (React Native + Expo, Node/Express, PostgreSQL) that turns parking enforcement into a crowdsourced early-warning system, alerting nearby drivers the moment their tires get chalked
- Designed a geospatial notification engine using **PostGIS ST_DWithin queries** that matches new chalk reports against active parking sessions and tiers alerts by proximity to the user’s street
- Built intersection-based location selection on the **OpenStreetMap Overpass API**, resolving street + cross-street pairs to precise coordinates and fixing a GPS-drift bug in the original design
- Migrated the original native iOS app to a cross-platform **Expo codebase** with JWT auth, persisted sessions, and a notification pipeline that never blocks the primary write path on failure

Watchlist Database

- Built a full-stack **Flask web app** backed by a normalized SQLite schema (users, lists, many-to-many links) for tracking personal movie, book, and reading/watching lists
- Integrated the **Open Library and TMDb APIs** so users can search real titles and add them with cover art, posters, authors, and directors to a catalog that grows from what users discover
- Implemented user accounts with hashed passwords and a normalized schema supporting many-to-many user-to-list relationships

Weather App

- Built **HTTP Basic Auth middleware** with role-based access control, gating an admin user-management dashboard behind an “admin” role
- Integrated the **OpenWeatherMap API** with a resilient request pipeline that handles network and parse errors so a failed lookup degrades gracefully instead of crashing the server
- Hardened the data layer by migrating to **better-sqlite3** with parameterized queries and gave the app a responsive glassmorphic redesign

EXPERIENCE

TD Canada Trust | Ontario

March 2026 - Present

Customer Experience Associate

- Process a high volume of deposits, withdrawals, bill payments, and transfers daily with accuracy, in accordance with banking regulations and TD’s Code of Conduct
- Assess individual customer needs to recommend tailored banking products and solutions, consistently meeting referral and sales objectives
- Resolve customer concerns using standard procedures, escalating non-standard issues to maintain a positive customer experience
- Collaborate with branch team members to deliver consistent, high-quality service and support a positive, inclusive work environment

Honda of Canada Mfg. | Alliston, ON

May - Sept. 2023, 2024, 2025

Summer Associate

- Assembled 2023-2025 Honda CRVs using state-of-the-art technology and computer-calibrated tools
- Repaired vehicles found during sweep, limiting outflow and maintaining quality for customers
- Handled power tools and developed correct posture for safely using them, resulting in safer work conditions